

TIC 238618726

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 3.166692 d
Epoch (t0) 1494.5954 BTJD
Duration 0.1700 d
Depth 11.824 % (118,235 ppm)
SNR 374.2

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

Eclipses observed 19 (gate ≥ 3)
Robust depth ratio 1.02 (gate ≥ 0.50)
Robust depth sigma 259.3 (gate ≥ 7.0)
Odd/even z-score 0.28 (consistent)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 9 light-curve dip in this sector: 176.5 sigma t0 refined by 0.0000 phase
Difference-image peak SNR 45.08 (gate ≥ 5.0) Compactness 0.90 (gate ≥ 0.50)
Dimming centroid offset 0.09 px (2 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 238618737
1.03 px away, dTmag +0.25 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)

